

Artificial Intelligence 2: Applications of Information Theory to Machine Learning

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Outline of Topics

- 1 Entropy and decision tree learning
- 2 Mutual Information for feature selection
- 3 Maximum Entropy Principle

Decision tree learning

Decision trees are one of the most popular machine learning algorithms because of their:

- Simplicity
- Interpretability: human can read and use decision tree. [Read this article.](#)
- Extendability: [Gradient boosting such as XGBoost](#) and [Random forest](#)

Applications:

- [Decision Trees for Credit Card Fraud Detection](#): *"It then creates an if/then decision tree for that account of features that do and don't point to fraud."*
- [Google Purchases Customer Service Automation Firm Onward](#):
"Onward uses a unique decision tree algorithm that allows its bots to be exceptionally functional in delivering world-class customer care."
- [AI Reveals The Perfect Pancake Recipe.](#)

Decision tree learning

Predictive model: a tree structure which consists of:

- Root or internal node: a independent variable (aka. a feature or an attribute)
- Leaf: an outcome of the depedent variable
- Branch: a rule (or decision)

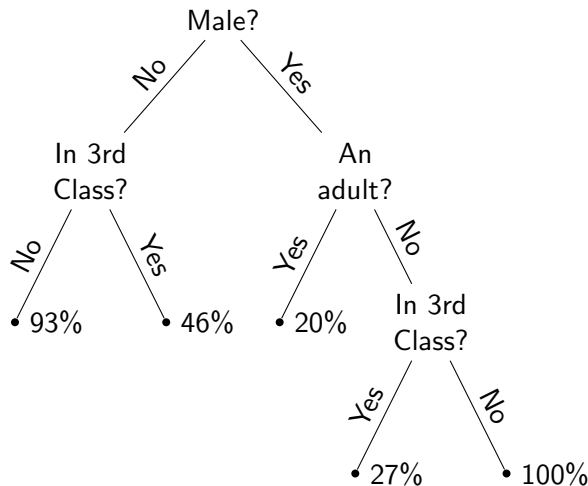
Types:

- Classification trees: dependent variables are categorical or qualitative (e.g. male/female).
- Regression trees: dependent variables are continuous or quantitative (e.g., Temperature)

Popular algorithms:

- ID3 (Iterative Dichotomiser 3)
- C4.5 (successor of ID3)
- CART (Classification And Regression Tree)

Decision tree example: survival of Titanic passengers

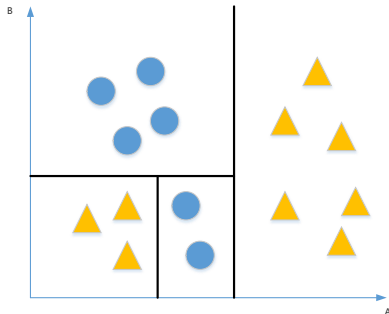


Titanic - Machine Learning from Disaster

Entropy for decision tree

Question: how to construct a decision tree? How to learn the structure from data?

Answer: given a data set, the algorithm groups and labels samples that are similar between them, and look for the best rules that split the samples that are dissimilar between them until they reach certain degree of similarity.



Gini index and Information gain

Question: How to look for the best rules that split the samples?

Answer: two main methods:

- Gini index (aka Gini impurity): the simplest one, used in [CART \(Classification And Regression Tree\)](#)
- **Information gain** ✓: used in
 - ID3 (Iterative Dichotomiser 3)
 - C4.5 (successor of ID3)

Gini index: Give a training dataset of J classes, it is defined as

$$I_G(p) = 1 - \sum_{i=1}^J p_i^2,$$

where p_i is the fraction of items labeled with class i in the dataset. For more details, please read this Nature Methods 2017 paper: [CART \(Classification And Regression Tree\)](#)

Information gain for decision tree

Information gain: the information we can gain after spiting the samples based on a independent variable (internal node)

Formally, information gain is defined as the change in information entropy H from a prior state to a state that takes some information as given:

$$IG(Y, X) = H(Y) - H(Y|X),$$

where Y is a random variable that represent the dependent variable (labels) and X is one of the independent variable, and $H(Y|X)$ is the conditional entropy of Y given X .

Example: Cancer diagnosis

We took some biopsy of some tissues of 5 patients who are suspected to have breast cancer. We measured three biomarkers, which are either positive (T) or negative (F). 3 out of these 5 patients are confirmed as cancer by oncologists, and labelled as 'T'. The three measures and labels are a set of training samples which are tabulated below:

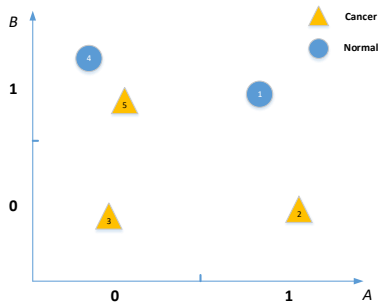
Tissue ID	Biomarker A	Biomarker B	Biomarker C	Label (Cancer?)
1	T	T	T	F
2	T	F	T	T
3	F	F	T	T
4	F	T	T	F
5	F	T	F	T

Question: How to train a decision tree? Or more precisely, how to split the samples by independent variables?

Example: Cancer diagnosis

Solution: We first use a random variable Y to model the dependent variable (i.e., label) with the range as $R_Y = \{0, 1\}$ and 0 means normal and 1 means cancer, and each of the independent variables, i.e., the biomarkers is a random variable, denoted as A , B and C , with range $R_A = R_B = R_C = \{0, 1\}$, respectively. Then we follow the following steps:

- Step 1: Calculate the entropy of random variable Y before the split



Example: Cancer diagnosis

Step 1: Calculate the entropy of random variable Y before the split

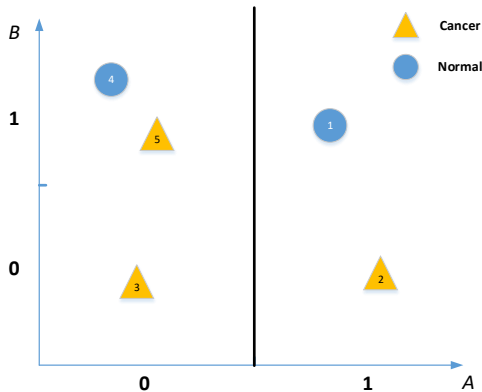
$$\begin{aligned} H(Y) &\equiv -E[\log P_Y(y)] \equiv -\sum_i^n P(Y = y_i) \log P(Y = y_i) \\ &\equiv -\sum_i^n p(y_i) \log p(y_i) \end{aligned}$$

$p(0) = \frac{2}{5}$ and $p(1) = \frac{3}{5}$, we have

$$\begin{aligned} H(Y) &= -[p(0) \log p(0) + p(1) \log p(1)] \\ &= -\left[\frac{2}{5} \log \frac{2}{5} + \frac{3}{5} \log \frac{3}{5}\right] = 0.971 \end{aligned}$$

Example: Cancer diagnosis

Step 2: Calculate the conditional entropy after a split. Since only three independent variables, we use brutal force, i.e., go over all three biomarkers:



Example: Cancer diagnosis

Step 2: Calculate the conditional entropy after the split. Since only three independent variables, we use brutal force, i.e., go over all three biomarkers and starting with A :

$$\begin{aligned} H(Y|A) &\equiv \sum_{a_i \in R_A} P_A(A = a_i) H(Y|A = a_i) \\ &= - \sum_{a_i \in R_A} P_A(A = a_i) \sum_{y_j \in R_Y} p(y_j|a_i) \log p(y_j|a_i) \end{aligned}$$

Since $P_A(A = 0) = \frac{3}{5}$, $P_Y(Y = 1|A = 0) = \frac{2}{3}$, $P_Y(Y = 0|A = 0) = \frac{1}{3}$, $P_A(A = 1) = \frac{2}{5}$, $P_Y(Y = 1|A = 1) = \frac{1}{2}$, and $P_Y(Y = 0|A = 1) = \frac{1}{2}$, we have

$$H(Y|A) = - \left[\frac{3}{5} \left(\frac{2}{3} \log \frac{2}{3} + \frac{1}{3} \log \frac{1}{3} \right) + \frac{2}{5} \left(\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right) \right] = 0.951$$

Example: Cancer diagnosis

Step 3: Calculate the information gain for split A :

$$IG(Y, A) = H(Y) - H(Y|A) = 0.951 - 0.971 = 0.02$$

Step 4: Repeat Step 2-3 to calculate the information gain of all splits.

Results:

$$\text{Split 1: } IG(Y, A) = 0.02$$

$$\text{Split 2: } IG(Y, B) = 0.419$$

$$\text{Split 3: } IG(Y, C) = 0.171$$

Candidate Split 2 has the highest information gain, so it will be the most favourable split for the root node.

Your first decision tree

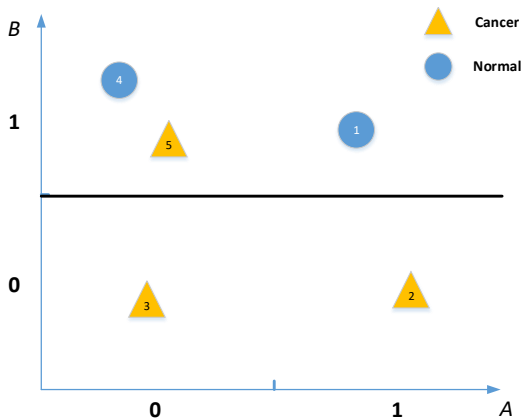
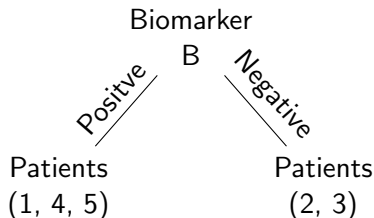


Figure 1: Splitting samples based on the values of random variable B generates the largest information gain.

Your first decision tree



Note: This example is oversimplified, e.g., the decision rules for splitting of samples, i.e., the branches, are directly based on the value (Positive/Negative) of the independent variable, because it is a binary discrete random variable as same as the dependent variable. However, for general discrete random variables with more than two values (called categorical variables) and continuous random variables, we need to search the best value, called cut-off, cut-point or threshold that maximise the information gain.

Drawbacks of decision trees

Drawbacks:

- Unstable: a small change in the data can lead to a large change in the structure of the optimal decision tree.
- Relatively inaccurate: Many other predictors such as Support Vector Machine and Neural Networks perform better than decision trees with similar data.

Solutions: Decision tree ensembles:

- Random forest
- Gradient boosting such as XGBoost

See [An Introduction to Statistical Learning](#): Chapters 8.1-8.2

Questions: Can we go beyond decision trees?

Information gain actually is...

Question: look at the equation of information gain:

$$IG(Y, X) = H(Y) - H(Y|X),$$

what do you remember?

Information gain actually is...

Answer: Mutual Information

$$I(X; Y) \equiv H(X) - H(X|Y) \quad (1)$$

$$\equiv H(Y) - H(Y|X) \quad (2)$$

$$\equiv H(X) + H(Y) - H(Y, X) \quad (3)$$

$$\equiv H(X, Y) - H(Y|X) - H(X|Y) \quad (4)$$

$$= D_{\text{KL}}(P(X, Y) \| P(X)P(Y)) \quad (5)$$

Interpretation: decision tree learning algorithms (ID3 and C4.5) recursively use mutual information to select the independent variable that share the most information with the dependent variable, then split (make decision) the samples based on the value of this independent variable.

Mutual Information Feature Selection

Question: Why not select a set of most relevant independent variables based on mutual information and use any other powerful machine learning algorithms to build predicted models?

Mutual Information feature selection: Use mutual information to choose a optimal set of independent variables, called features that allow us to classify samples. Formally, given an initial set F with n independent variables, $\mathbf{X} = \{X_1, X_2 \dots, X_n\}$, find subset with $S \subset F$ features that maximises the Mutual Information $I(Y; S)$ between the dependent variable Y (label), and the subset of selected features S .

- [Wikipedia: Feature selection](#)
- [Scikit-learn: Feature Selection](#)

Mutual Information Feature Selection

Mutual information feature selection

Initialisation: Set $F \leftarrow \mathbf{X}$ and $S \leftarrow \emptyset$

$$f_{\max} = \operatorname{argmax}_{X_i \in \mathbf{X}} I(Y; X_i)$$

Set $F \leftarrow F \setminus \{f_{\max}\}$ and $S \leftarrow f_{\max}$

Repeat until $|S| = K$:

$$f_{\max} = \operatorname{argmax}_{X_i \in F} (I(Y; X_i) - \beta \sum_{X_s \in S} I(X_s; X_i))$$

Set $F \leftarrow F \setminus \{f_{\max}\}$ and $S \leftarrow f_{\max}$

End

End

⁰R. Battiti, "Using mutual information for selecting features in supervised neural net learning," **IEEE Trans. on Neural Networks**, 5(4), 1994.

Mutual information feature selection

Further readings:

- Decision trees: from efficient prediction to responsible AI, *Front. Artif. Intell.*, July 2023
- Normalized Mutual Information Feature Selection, *IEEE Trans. on Neural Networks*, 20(2), 2009.

Maximum Entropy Principle

Maximum Entropy Principle:

- Jaynes' Maximum Entropy Principle
- Duality of Maximum Likelihood and Maximum Entropy

Fundamental question: How information, energy and matter, the three fundamental elements of our universe, give rise to life and consciousness?

Probably **Non equilibrium thermodynamics**, which inspired diffusion probabilistic models, e.g., **Stable Diffusion** might offer some clues.

Exercise

Task: After learning how to implement decision tree learning algorithm to predict Titanic passenger survival, implement the mutual information feature selection algorithm and construct a decision tree. Compare it with the decision tree learned from data without feature selection.